Heart Attack Prediction Using Machine Learning Models

Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
IQR	Interquartile range
HGBC	Histogram-based Gradient Boosting Classifier
SMOTE	Synthetic Minority Oversampling Technique

1. Introduction

Heart attacks are life-threatening. The goal of this project is to identify the best machine learning model for heart attack prediction and further improve it.

2. Dataset Description and Analysis

'heart_attack.csv' contains fifteen features. The dataset has 129,998 records. Table 1 describes features in the dataset.

Table 1

Dataset Features Description

Feature	Description
HighBP	Does patient have high blood pressure?
HighChol	Does patient have high cholesterol levels?
CholCheck	Has patient done a cholesterol screening check?
BMI	Body mass index (this is a continuous variable)
Smoker	Does patient smoke?
Stroke	Had patient had a stroke before?
Diabetes	What diabetic level is patient in? (ordinal variable)
PhysActivity	Does patient do any physical activity
HvyAlcoholConsump	Does patient drink heavily?
MentHlth	How many days a month does patient have poor mental health?
PhysHlth	How many days a month does patient have poor physical health?
Sex	Is the patient male or female?
Age	What age group does patient belong to? (ordinal variable)
Education	What level of education has patient attained? (ordinal variable)
Income	How much does patient earn a year (interval variable)

Exploratory data analysis was conducted to understand the distribution of variables. Key observations:

- Only 9.35% of cases were positive, indicating a class imbalance
- 96% of individuals underwent cholesterol screening
- 76% of individuals are physically active
- 5.7% of individuals are heavy alcohol consumers
- Average age group: 60–64 (group 8)

- ‘BMI’, ‘Diabetes’, ‘MentHlth’, ‘PhysHlth’, ‘Age’, ‘Education’, and ‘Income’ are not binary values
- Maximum ‘BMI’ was 98 (likely an outlier)
- ‘BMI’ had ten null values

Figure 2.1 demonstrates the imbalanced distribution of patients with a heart attack and those without. Class imbalance negatively affects the performance of models and will be handled during data pre-processing (Tran et al., 2021).

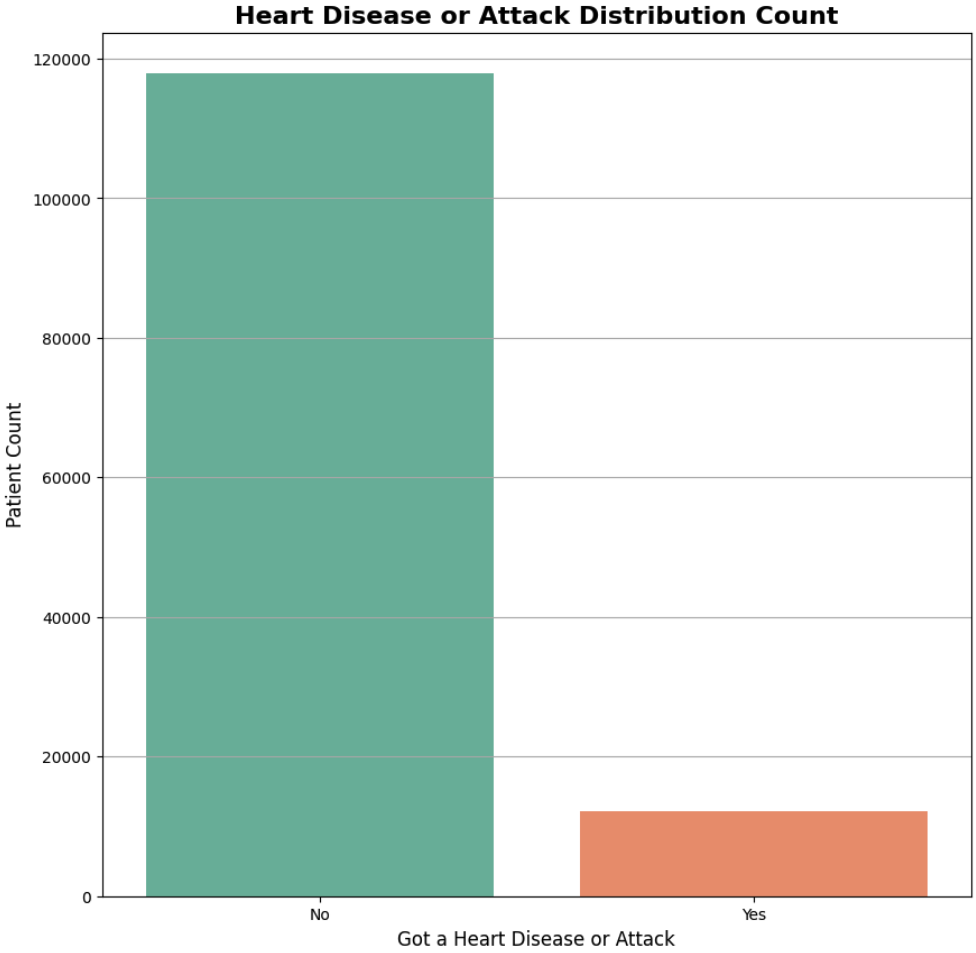

Figure 2.1: Target variable distribution

A correlation heatmap was generated to analyse relationships between variables (Figure 2.2). The correlation values range from -1 (strong negative relationship) to 1 (strong positive relationship) between variables.

Key relationships:

- ‘HighBP’ showed a positive correlation with ‘Age’ (0.34), ‘HighChol’ (0.30), and ‘Diabetes’ (0.27).
- ‘PhysHlth’ showed a positive correlation with ‘MentHlth’ (0.36), indicating patients with poor mental health may also experience poor physical health.
- ‘Education’ and ‘Income’ had the highest correlation (0.45)

Most correlations between features are weak, indicating that no single feature dominated the dataset, hence combining multiple features is likely more effective for prediction.

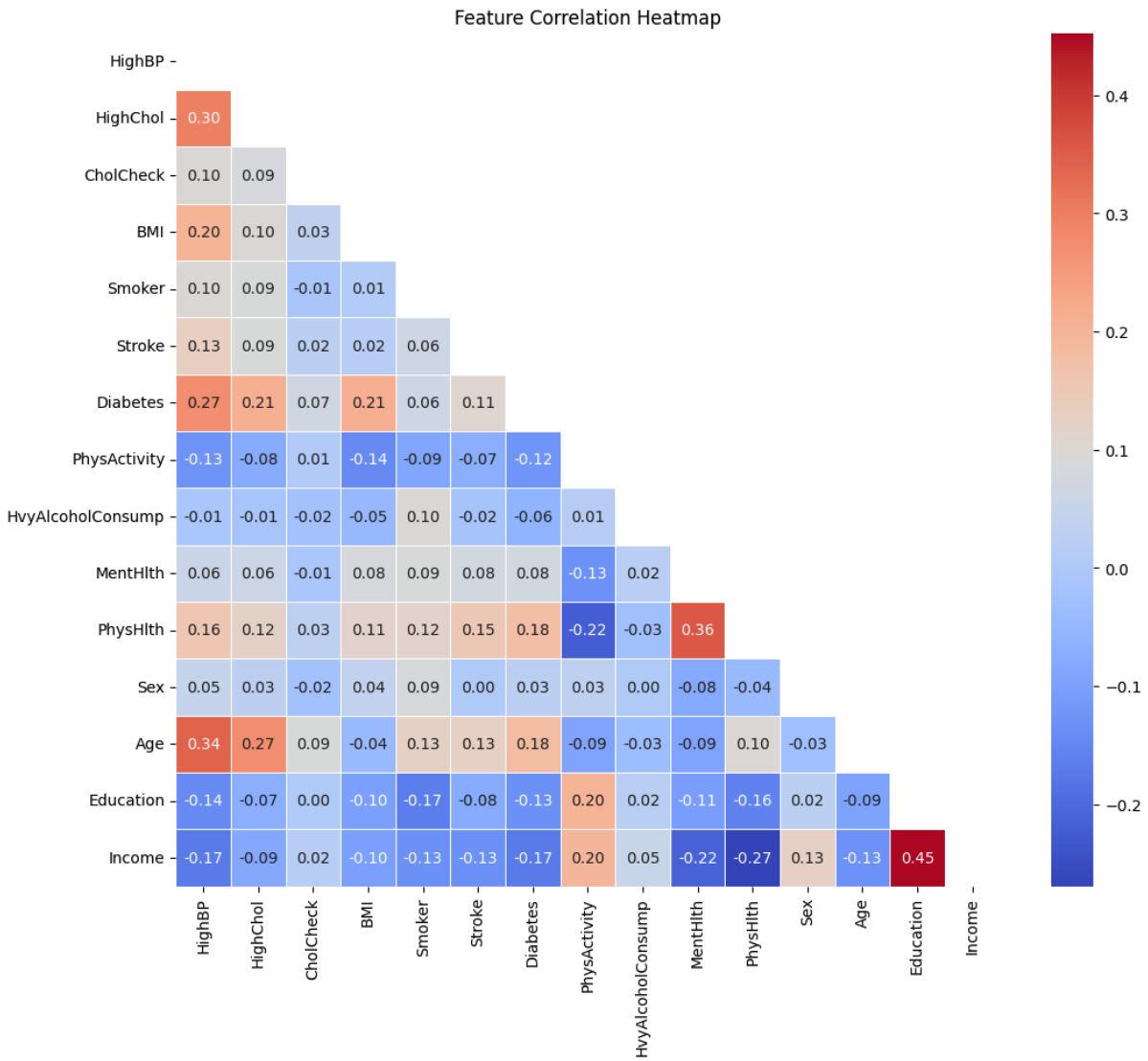

Figure 2.2: Feature correlation heatmap

To understand the influence of individual features on the target variable, the correlation between each feature and the target variable was analyzed, as shown in Figure 2.3.

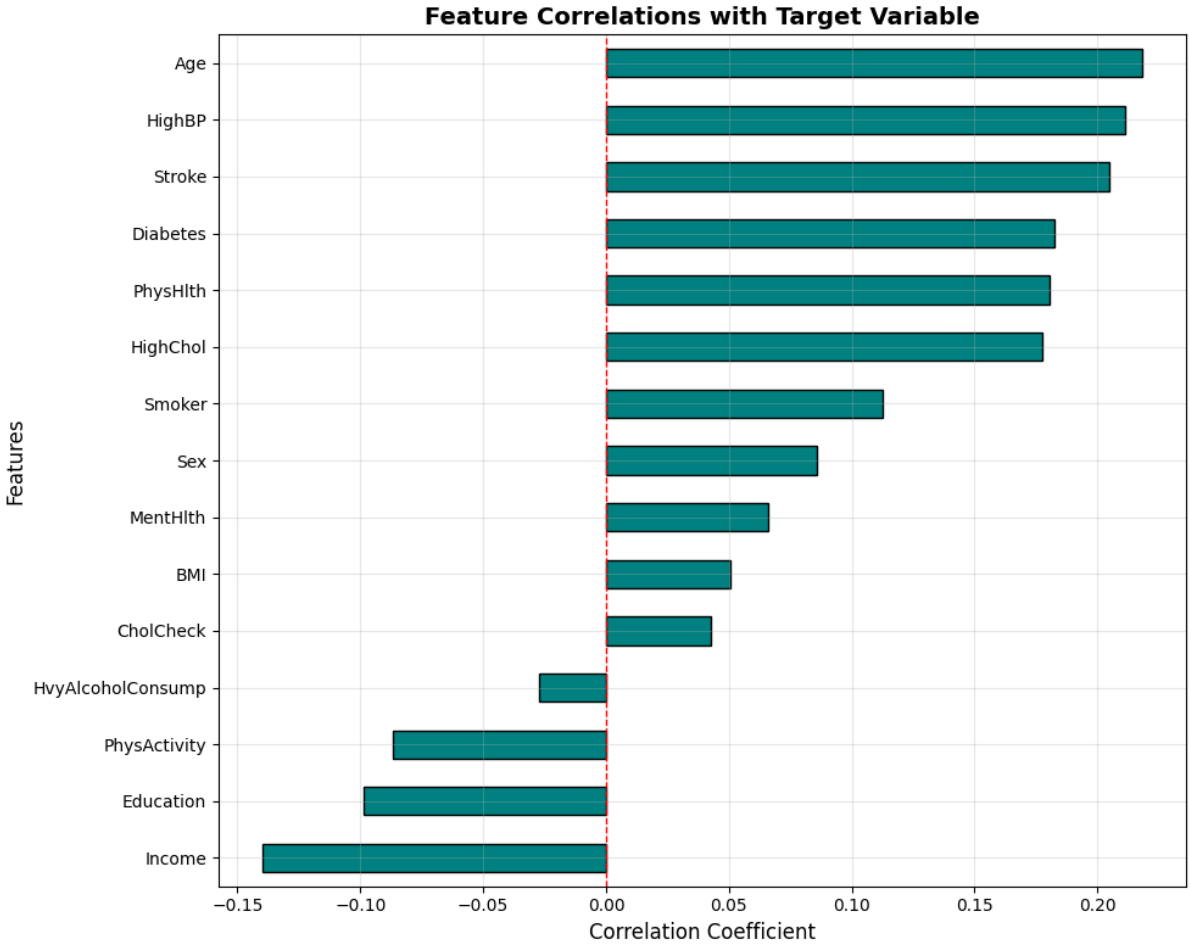

Figure 2.3: Correlation of features with the target feature

'Age' (0.22), 'HighBP' (0.21), 'Stroke' (0.20), 'Diabetes' (0.18), 'PhysHlth' (0.17) and 'HighChol' (0.17) showed the highest positive correlations. 'Smoker', 'Sex', 'MentHlth', 'BMI', and 'CholCheck' demonstrated moderate correlations. 'HvyAlcoholConsump' (-0.03), 'PhysActivity' (-0.09), Education (-0.10), and Income (-0.14) showed inverse relationships with the target feature. Overall, no feature was strongly predictive on its own.

3. Data Preparation and Pre-Processing

3.1. Handling Missing Values

Ten data rows had missing values on the 'BMI' feature and were dropped.

	Null Value Count
HeartDiseaseorAttack	0
HighBP	0
HighChol	0
CholCheck	0
BMI	10
Smoker	0
Stroke	0
Diabetes	0
PhysActivity	0
HvyAlcoholConsump	0
MentHlth	0
PhysHlth	0
Sex	0
Age	0
Education	0

Figure 3.1.1: Missing values summary

3.2. Dataset Splitting

The dataset was split into training and testing subsets. A 7:3 ratio was used to split the data; 70% training data and 30% testing data.

3.3. Outlier Detection

Outlier analysis was conducted using boxplot and the IQR method, as shown in Figure 3.3.1 and 3.3.2. 'BMI' was the only continuous feature with potential outlier data points, with the maximum value reaching 98, with mean 28.44 and quartiles (Q1=24.0, Q2=27.0, Q3=31.0). The boxplot shows many outliers in the 'BMI' distribution, especially on the higher end of the scale. Values outside the IQR bounds were capped to reduce the influence of outliers while preserving the dataset size.

	count	mean	std	min	25%	50%	75%	max
HeartDiseaseorAttack	129998.0	0.093509	0.291146	0.0	0.0	0.0	0.0	1.0
HighBP	129998.0	0.429614	0.495023	0.0	0.0	0.0	1.0	1.0
HighChol	129998.0	0.426322	0.494544	0.0	0.0	0.0	1.0	1.0
CholCheck	129998.0	0.963315	0.187988	0.0	1.0	1.0	1.0	1.0
BMI	129988.0	28.435586	6.999582	12.0	24.0	27.0	31.0	98.0
Smoker	129998.0	0.448030	0.497294	0.0	0.0	0.0	1.0	1.0
Stroke	129998.0	0.041354	0.199110	0.0	0.0	0.0	0.0	1.0
Diabetes	129998.0	0.296689	0.697608	0.0	0.0	0.0	0.0	2.0
PhysActivity	129998.0	0.760758	0.426622	0.0	1.0	1.0	1.0	1.0
HvyAlcoholConsump	129998.0	0.057562	0.232915	0.0	0.0	0.0	0.0	1.0
MentHlth	129998.0	3.159718	7.344179	0.0	0.0	0.0	2.0	30.0
PhysHlth	129998.0	4.222942	8.689544	0.0	0.0	0.0	3.0	30.0
Sex	129998.0	0.439214	0.496293	0.0	0.0	0.0	1.0	1.0
Age	129998.0	8.056624	3.048218	1.0	6.0	8.0	10.0	13.0
Education	129998.0	5.078670	0.977537	1.0	4.0	5.0	6.0	6.0
Income	129998.0	6.098894	2.057443	1.0	5.0	7.0	8.0	8.0

Figure 3.3.1: Statistical distribution of features in the dataset

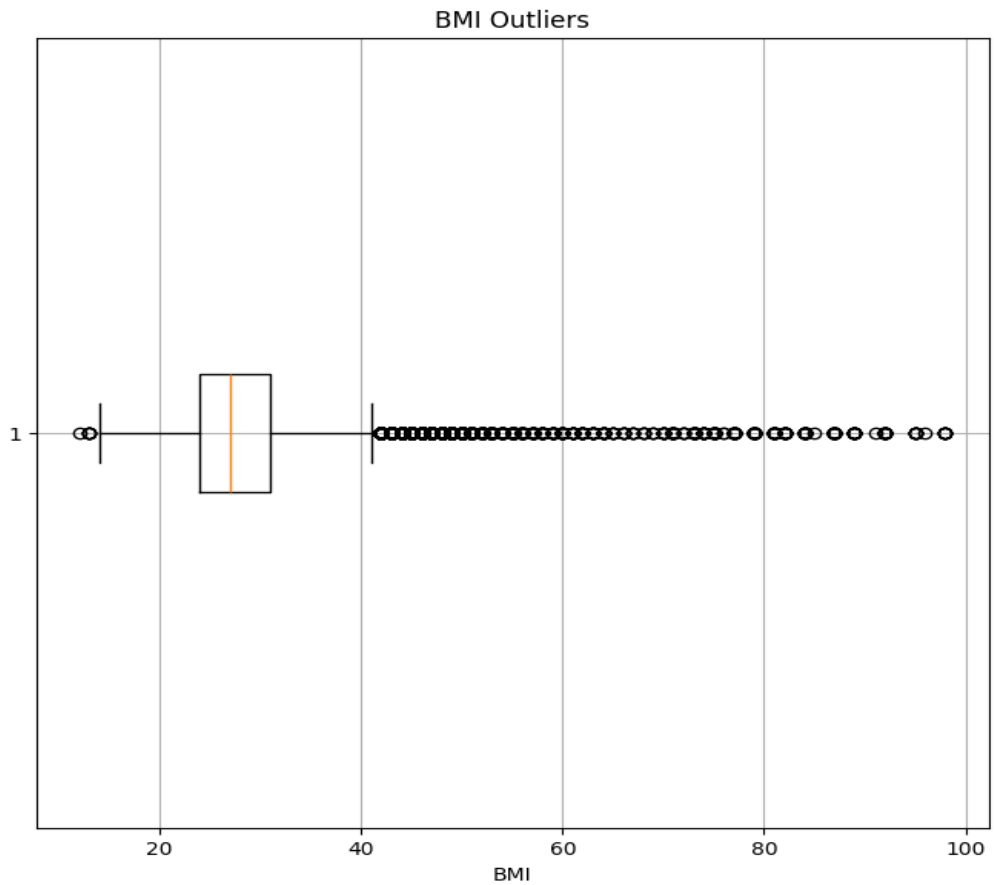

Figure 3.3.2: Boxplot showing outliers in BMI distribution

3.4. Feature Scaling

Numerical features were standardised using the StandardScaler to ensure features with larger numeric ranges did not dominate model training. The scaler was fitted using training data and subsequently applied to the testing subset to prevent data leakage.

3.5. Data Augmentation

As observed in Figure 3.5.1, the dataset was imbalanced as only 9.35% of the cases were positive. SMOTE was used to balance the dataset and reduce bias by generating synthetic samples for the minority class (Mohammed et al., 2020).
Augmentation was applied to the training dataset after splitting to avoid data leakage.
Figure 3.5.1 shows the distribution of data before and after applying SMOTE.

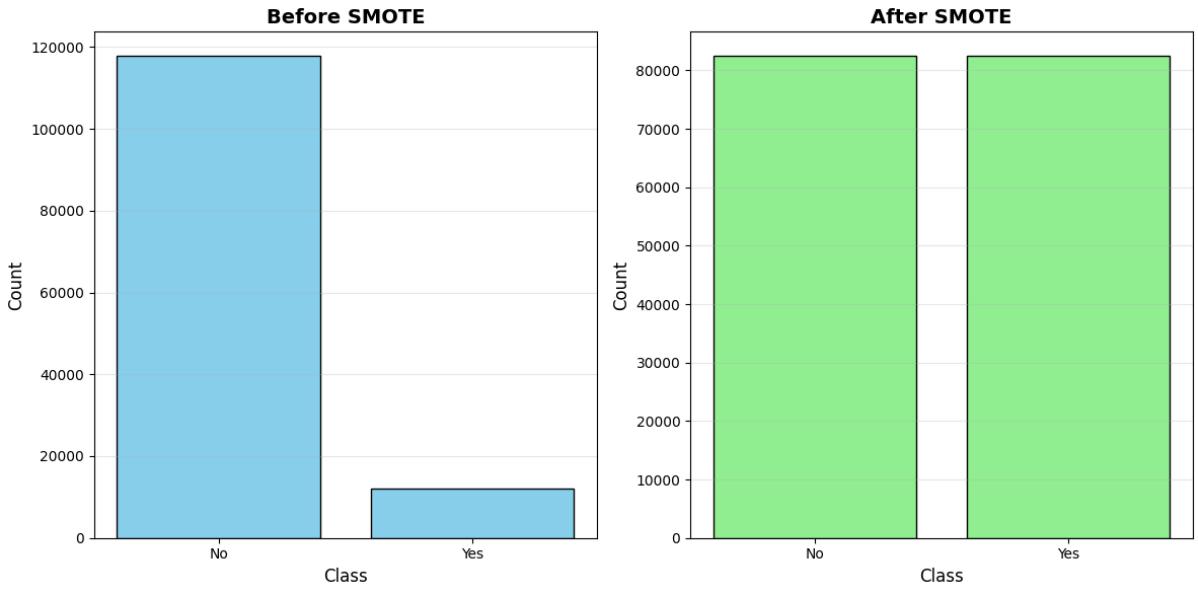

Figure 3.5.1: Distribution before and after applying SMOTE

4. Model Selection & Training

4.1. Model Selection

Six classification machine learning models were trained, namely;

- Logistic regression
- K-Nearest Neighbor
- Decision Tree
- Random Forest
- Artificial Neural Network
- Histogram-based Gradient Boosting Classifier(HGBC)

4.2. Model Training

All selected models were trained using the preprocessed dataset. Each model was fitted using its respective learning algorithm. Training enabled each model to learn patterns between features and the target variable.

5. Model Evaluation

5.1. Evaluation Metrics

Evaluation metrics used;

- F1-score
- Accuracy
- Recall
- Precision
- Specificity
- AUC

5.2. Overall Model Performance

Model	f1_score	accuracy	recall	precision	specificity	auc
MLP	0.0929	0.9084	0.0501	0.6310	0.9970	0.8378
Logistic Regression	0.1673	0.9078	0.0989	0.5420	0.9914	0.8372
HistGradientBoosting (0.3 threshold)	0.3846	0.8121	0.6274	0.2773	0.8312	0.8305
KNN	0.2593	0.8487	0.2830	0.2393	0.9071	0.6570
Random Forest	0.1068	0.8434	0.1000	0.1145	0.9202	0.5374
Decision Tree	0.1385	0.4964	0.4323	0.0824	0.5031	0.4677

Figure 5.2.1: Overall performance comparison of the models

5.3. Class 1 (Heart Attack) Model Performance

Model	class_1_precision	class_1_recall	class_1_f1	class_1_support
MLP	0.6310	0.0501	0.0929	3650.0
Logistic Regression	0.5420	0.0989	0.1673	3650.0
HistGradientBoosting (0.3 threshold)	0.2826	0.6134	0.3870	3650.0
KNN	0.2391	0.2830	0.2592	3650.0
Random Forest	0.0525	0.0307	0.0387	3650.0
Decision Tree	0.0861	0.4178	0.1428	3650.0

Figure 5.3.1: Heart attack cases (Class 1) prediction performance comparison of the models

5.4. Class 0 (No Heart Attack) Model Performance

Model Name	class_0_precision	class_0_recall	class_0_f1	class_0_support
MLP	0.9104	0.9970	0.9517	35347.0
Logistic Regression	0.9142	0.9914	0.9512	35347.0
HistGradientBoosting (0.3 threshold)	0.9546	0.8392	0.8932	35347.0
KNN	0.9245	0.9070	0.9157	35347.0
Random Forest	0.9040	0.9428	0.9230	35347.0
Decision Tree	0.9002	0.5423	0.6769	35347.0

Figure 5.4.1: No-heart attack cases (Class 0) prediction performance comparison of the models

5.5. Discussion of Results

The Multilayer Perceptron (MLP) achieved the highest accuracy (0.9084) and AUC (0.8378). However, analysis of class 1 performance showed it recorded a low recall (0.0501). This indicates bias towards the majority class. A class 0 recall score of 0.9970 supported this argument.

The K-Nearest Neighbors (KNN) model showed biased performance across both classes, with a recall score of 0.2797 for class 1 and 0.9070 for class 0. Random Forest and Decision Tree performed poorly, particularly in detecting positive cases. Both models had a low recall for class 1.

The HGBC model provided the best balance between overall and class-specific performance. It achieved a high recall for class 1 (0.6134), indicating capability in detecting positive cases, while maintaining an accuracy score of 0.8113 and AUC score of 0.8314. The model's class 0 performance remained strong with a recall score of 0.8392, showing balanced detection of both classes. This makes it the most appropriate model for this task.

These results show the limitation of relying solely on accuracy with imbalanced data, as models may achieve high accuracy by predominantly predicting the majority class, while failing to detect important positive cases.

6. Model Comparison

Figure 6.1 shows the overall performance comparison of all models

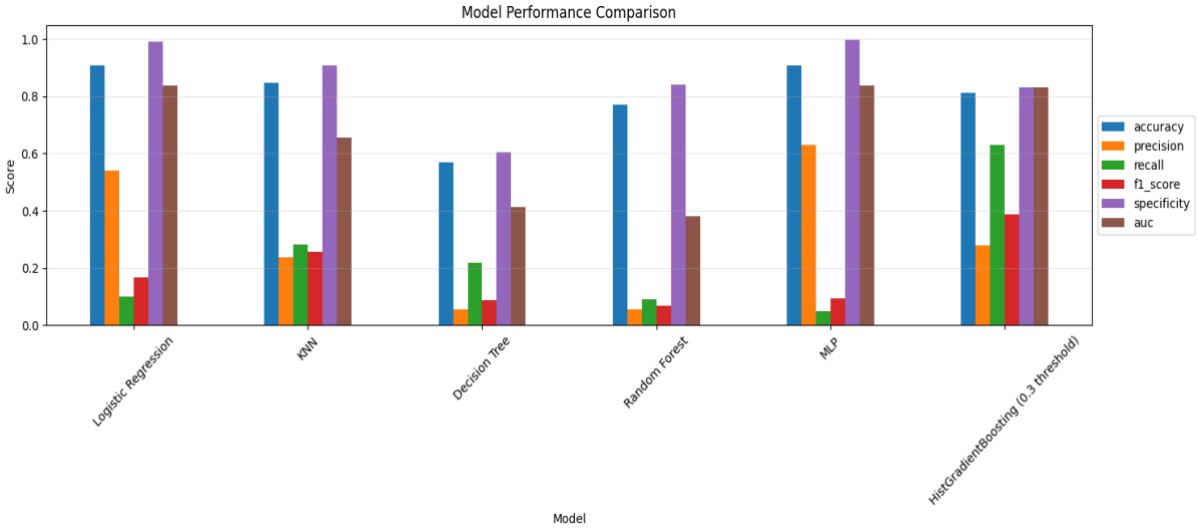

Figure 6.1: Overall performance comparison of all models

Recall was the most suitable evaluation metric. Models with higher recall were preferred to minimise false negatives, which can lead to delayed treatment or fatal outcomes. The Multilayer Perceptron achieved the highest accuracy, but had low recall, making it unsuitable for this task. The HGBC model provided the best balance between performance metrics. It had the highest recall while maintaining a competitive accuracy.

The KNN model achieved moderate performance but did not outperform other models in any key metric. Random Forest and Decision Tree models demonstrated poor ability to correctly identify positive cases.

The HGBC model was the most appropriate model for this task, as accurate identification of positive cases was prioritized over overall classification accuracy.

7. Improving Model Performance

7.1. Threshold Optimization

During training of the HGBC model, the default classification threshold (0.5) was adjusted to 0.3 through threshold optimization.

To further improve the recall score, a manual threshold analysis was conducted to analyse the effect of different thresholds. Thresholds (0.25, 0.30, and 0.35) were evaluated using model probabilities. The results showed that a decrease in the classification threshold improved recall while reducing precision.

A classification threshold of 0.25 resulted in the highest recall score of 0.71, meaning 71% of positive cases were correctly detected. Further reducing the classification threshold led to a sharp decline in precision and a large increase in false positives. Figure 7.1.1 shows how precision and recall varied as the classification threshold changed.

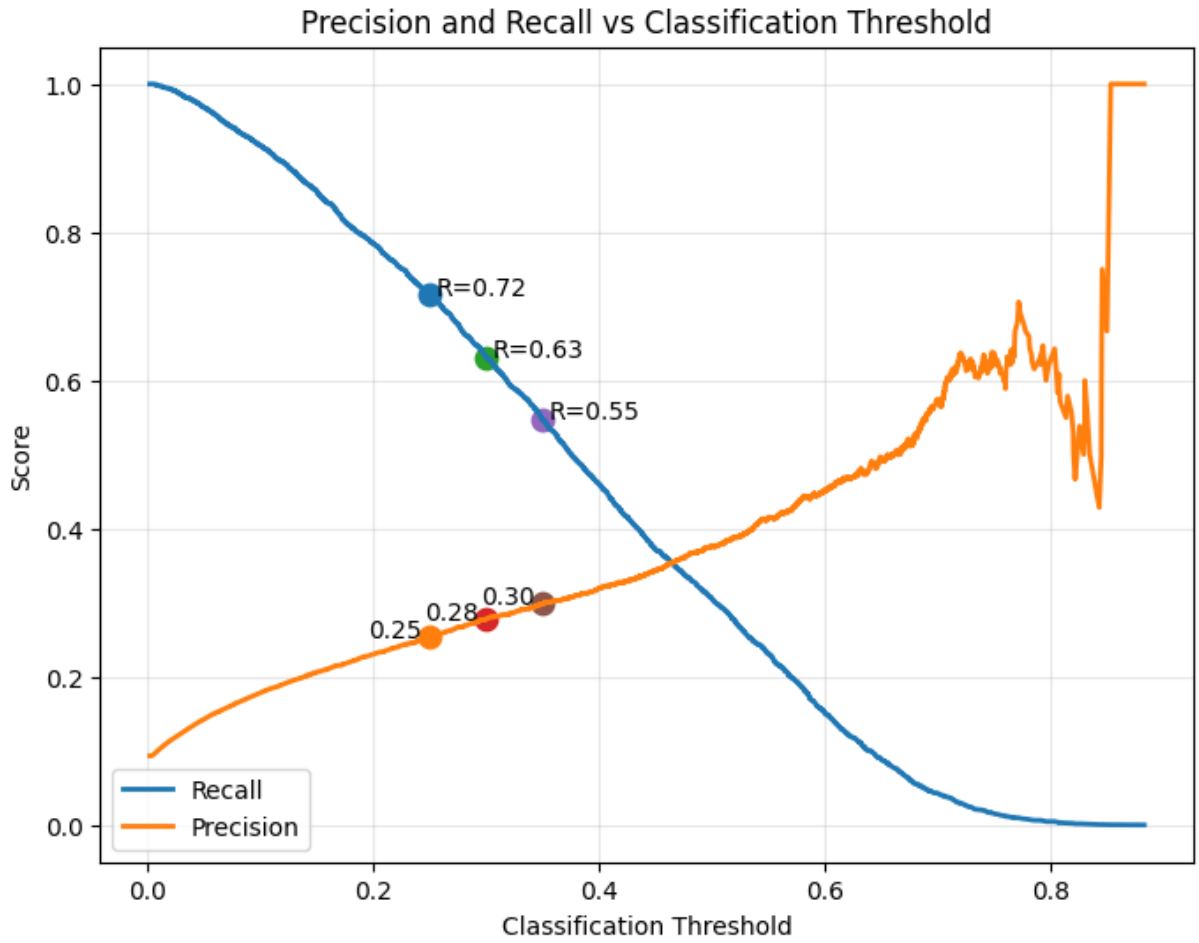

Figure 7.1.1: Precision and Recall vs Classification Threshold

Figures 7.1.2 and 7.1.3 illustrate the impact of lowering the classification threshold from 0.30 to 0.25. True positives increased from 2,338 to 2,593, showing improved detection of positive cases. False negatives decreased from 1,312 to 1,057, meaning fewer actual cases were incorrectly classified as negative cases. However, false positives rose from 6,199 to 7,813, meaning more individuals were incorrectly classified as having a heart attack. True negatives decreased from 29,148 to 27,534. Lowering the threshold improved recall but lowered precision.

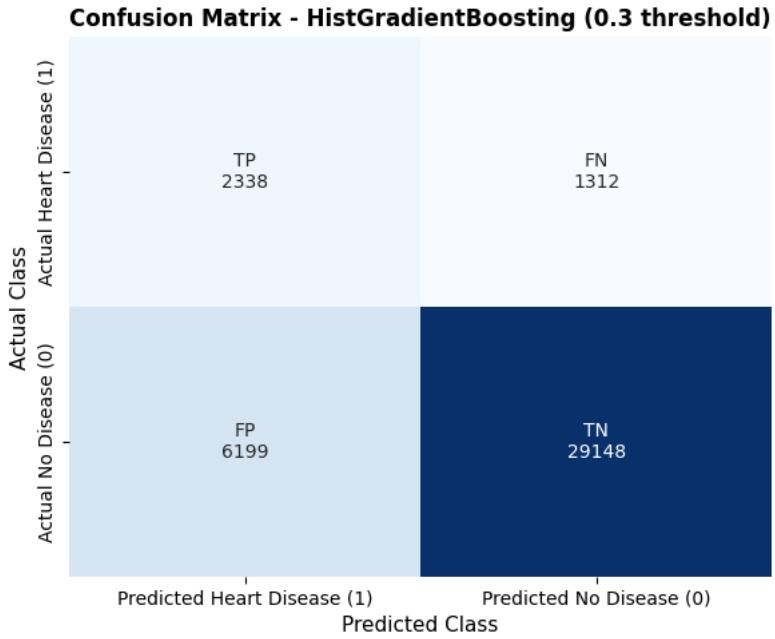

Figure 7.1.2: Confusion matrix for HGBC model at classification threshold 0.30

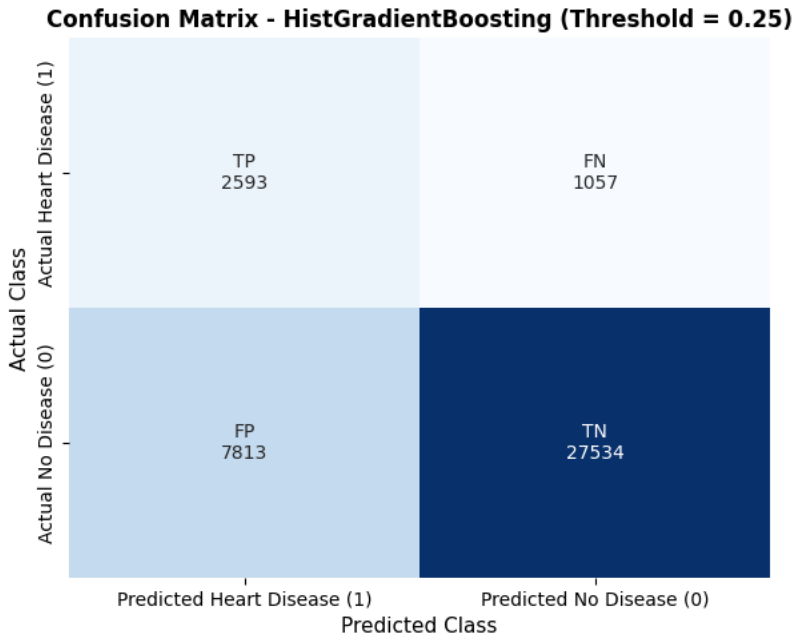

Figure 7.1.3: Confusion matrix for HGBC model at classification threshold 0.25

7.2. Cost-sensitive Learning through Sample Weights

Although SMOTE was used to balance the training data, the original dataset contained fewer positive heart attack cases, and the test set retained the natural class distribution. To improve detection of positive cases, higher weights were assigned to heart attack samples so that misclassifying these observations incurred a greater penalty. Figure 7.2.1 highlights the impact of applying cost-sensitive learning. True positives increased from 2,593 to 3,081, showing improved detection of positive cases. False negatives decreased from 1,057 to 569, meaning fewer actual cases were incorrectly classified as negative cases. However, false positives rose from 7,813 to 11,622, indicating more individuals had a heart attack, yet they did not. True negatives decreased from 27,534 to 23,725.

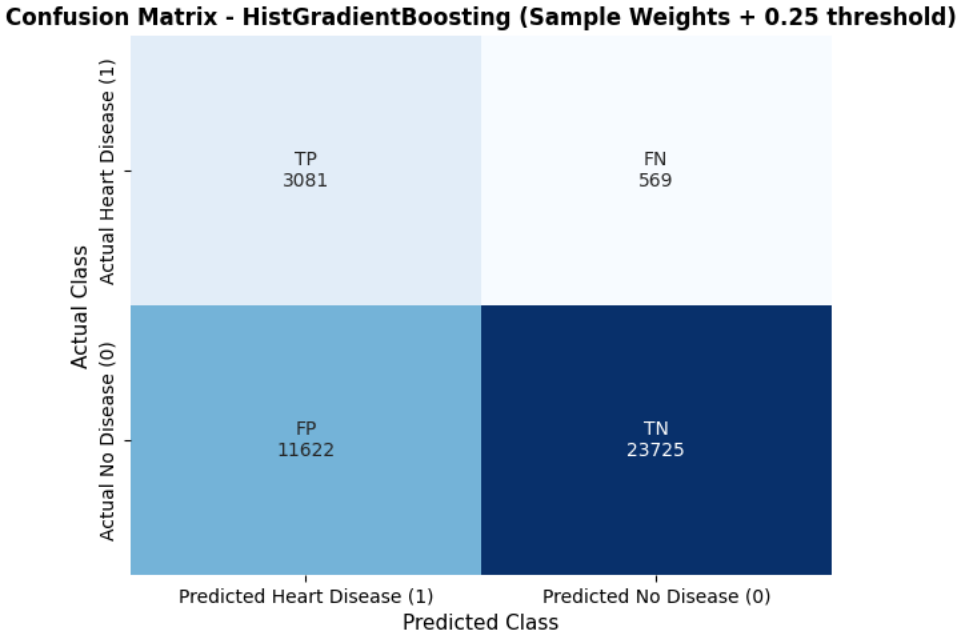

Figure 7.2.1: Confusion matrix for HGBC model with sample weighting and threshold=0.25

8. Conclusion

Both model improvement methods applied to the HGBC model had a positive impact on class one's recall score. Figure 8.1 shows class one's recall score improvement from 0.6405 to 0.7104 after threshold optimization, and an improvement from 0.7104 to 0.8441 after applying cost-sensitive learning.

Model Name	class_1_precision	class_1_recall	class_1_f1	class_1_support
HistGradientBoosting (Sample Weights + 0.25 th...	0.2095	0.8441	0.3357	3650.0
HistGradientBoosting (Threshold = 0.25)	0.2492	0.7104	0.3690	3650.0
HistGradientBoosting (0.3 threshold)	0.2739	0.6405	0.3837	3650.0

Figure 8.1: Class 1 recall metric improvement

Figure 8.2 shows class zero's recall score decrease from 0.8246 to 0.7790, and finally to 0.6712 which is relatively high.

Model	class_0_precision	class_0_recall	class_0_f1	class_0_support
HistGradientBoosting (Sample Weights + 0.25 th...	0.9766	0.6712	0.7956	35347.0
HistGradientBoosting (Threshold = 0.25)	0.9630	0.7790	0.8613	35347.0
HistGradientBoosting (0.3 threshold)	0.9569	0.8246	0.8859	35347.0

Figure 8.2: Class 0 recall metric improvement

References

Mohammed, A. J., Muhammed Hassan, M., & Hussein Kadir, D. (2020). Improving classification performance for a novel imbalanced medical dataset using SMOTE method. *International Journal of Advanced Trends in Computer Science and Engineering*, *9*(3), 3161-3172.

Tran, N., Chen, H., Jiang, J., Bhuyan, J., & Ding, J. (2021). Effect of class imbalance on the performance of machine learning-based network intrusion detection. *International Journal of Performability Engineering*, *17*(9), 741.